# LMS Algorithm

# Least Mean-Squares (LMS) algorithm

- The **Least Mean-Squares (LMS) algorithm** is a widely used adaptive filter technique in neural networks, signal processing, and control systems.
- Developed by Bernard Widrow and Ted Hoff in 1960, the LMS algorithm is a stochastic gradient descent method that iteratively updates filter coefficients to minimize the mean square error between the desired and actual signals.

# Least Mean-Squares (LMS) algorithm

- The Least Mean Squares (LMS) method is an adaptive algorithm widely used for finding the coefficients of a filter that will minimize the mean square error between the desired signal and the actual signal.
- It is mainly utilized in training algorithms such as gradient descent, where the network finalizes a target function by iteratively adjusting its weights w.r.t. the error between predicted and actual outputs.

# Key Concepts:

- **Adaptive Filtering**: Adaptive filters adjust their coefficients based on the input signal. The LMS algorithm is an example of an adaptive filter.
- **Mean Square Error (MSE):** This is the criterion the LMS algorithm aims to minimize. MSE is the expectation of the square of the error signal.
- **Error Signal (e(n)):** The difference between the desired signal (d(n)) and the output of the filter (y(n)). $e(n) = d(n) - x^T(n)w(n)$
- **Filter Coefficients (w(n)):** The parameters of the filter that are updated iteratively to minimize the MSE.

# Mathematical Foundation of LMS algorithm

LMS algorithm is based on using the immediate values of cost function $\varepsilon(w)$ which can be expressed in terms of error signal as:

$\varepsilon(w) = \frac{1}{2}e^2(n)$

where,

- e(n) is the error calculated by the difference between desired and target output values.
- $e^2(n)$ is used here because we are using squared error and $\frac{1}{2}$ is used for simplifying the calculation purpose.

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where

- d(n)-desired output value
- $x^T(n)$- transpose of input vector x
- w(n) - weight vector

**So, cost function can be written as:**

$\varepsilon(w) = \frac{1}{2}(d(n) - x^T(n)w(n))^2$

# Mathematical Foundation of LMS algorithm

differentiating the cost function $\varepsilon$ with respect to weight vector w:

- $\frac{\delta\varepsilon(w)}{\delta w(n)} = e(n)\frac{\delta e(n)}{\delta w(n)}$
- $\frac{\delta e(n)}{\delta w(n)} = -x(n)$
- So, $\frac{\delta\varepsilon(w)}{\delta w(n)} = -x(n)e(n) = g'(n)$

where g'(n) is the estimate of gradient vector

# LMS Algorithm in Steepest Descent Method

**Steepest descent method** is a general optimization technique used to find the minimum of a function. It iteratively updates the parameters in the direction of the negative gradient of cost function.

In this method, weight updating works as:

w(n+1) = w(n) - $\eta g(n)$ where $\eta$ is the learning rate parameter and g(n) is the gradient evaluated at vector point w(n)

$$\Delta w(n) = w(n+1) - w(n) = -\eta g(n)$$

***LMS algorithm is the specific implementation of the steepest descent method, mostly used for adaptive filtering and linear regression. In other words, we can say that LMS algorithm is the application of steepest descent.***

# LMS Algorithm in Steepest Descent Method

w'(n+1) = w'(n) - $\eta g'(n)$

Substituting the value of g'(n) calculated before, we get:

$$w'(n+1) = w'(n) + \eta x(n)e(n)$$

where,

- w'(n) is the estimate of weight vector.
- and $\eta$ is the learning rate parameter.

Now substituting the value of error signal e(n):

$$w'(n+1) = w'(n) + \eta x(n)[d(n) - x^T(n)w'(n)]$$

# LMS Algorithm in Steepest Descent Method

$$w'(n+1) = w'(n) + \eta x(n)[d(n) - x^T(n)w'(n)]$$

Taking w'(n) as common and solving the equation as:

$$w'(n+1) = [I - \eta x(n)x^T(n)]w'(n) + \eta x(n)d(n)$$

where,

- I is the Identity matrix, i.e. a square matrix containing 1 in its all diagonal values.
- $w'(n) = z^{-1}[w'(n+1)]$
- where $z^{-1}$ is a unit delay operator. It represents a delay of one time step.

# Convergence Consideration of LMS algorithm

- Convergence refers to the algorithm's ability to reach a steady state where the error signal becomes minimal. For the LMS algorithm, convergence behavior of LMS algorithm depends on the input vector $x(n)$ and learning rate parameter value $\eta$.